

Joohyung Lee

ML Researcher, AITRICS (Seoul, South Korea)



Website



LinkedIn



Scholar



GitHub

✉ joohyunglee.research@gmail.com

RESEARCH FOCUS & INTERESTS

- I build **structure-respecting learning systems** that reliably satisfy principled regularities under distribution shift. My approach translates regularities into scalable mechanisms: self-supervised objectives, soft regularizers, and architecture-level inductive biases.
- **Research interests (PhD):** (i) diagnosing and mitigating **invariance–equivariance interference** via subspace decoupling; (ii) **learning approximate symmetries** from spatiotemporal signals and enforcing them with layer-local constraints; (iii) **semantic/physical plausibility constraints** for trustworthy generative and world models.

EDUCATION

- **Texas A&M University** College Station, TX, USA
M.S. in Electrical Engineering Aug 2014
- **The University of Texas at Dallas** Richardson, TX, USA
B.S. in Electrical Engineering Aug 2012

RESEARCH & INDUSTRY EXPERIENCE

- **AITRICS** Seoul, South Korea
ML Researcher May 2022 – Present
 - **Geometric / structure-respecting SSL:** proposed *Soft Equivariance Regularization (SER)* to decouple invariant SSL at the final embedding from equivariance constraints at intermediate spatial feature maps (ICLR 2026, accepted; first author).
 - **Trustworthy multimodal EHR modeling:** developed self-supervised and foundation-model approaches for predicting critical patient events from irregular, partially observed clinical time series (MLHC 2023 spotlight; ICLR 2023 TML4H workshop, Best Paper Honorable Mention).
 - **Korea Electronics Technology Institute (KETI)** Seongnam-si, South Korea
Senior Researcher Dec 2020 – May 2022
 - Designed self-supervised pretraining and transfer-learning pipelines for medical image analysis under limited labels and distribution shift.
 - **KAIST, Robotics & Computer Vision Lab** Daejeon, South Korea
Researcher Mar 2020 – Dec 2020
 - Developed model architectures for anisotropic 3D medical volumes; contributed to volumetric rectal cancer staging (MICCAI 2022).
 - **National Cancer Center** Gyeonggi-do, South Korea
Researcher Oct 2016 – Mar 2020
 - Studied robustness and uncertainty of deep medical models in low-data regimes; published on variance reduction for segmentation (IEEE Access 2019).
 - **Laboratory for Optical Diagnosis & Imaging, Texas A&M University** College Station, TX, USA
Student Researcher Mar 2013 – Jun 2014
 - Developed classical ML pipelines (feature extraction/selection; supervised learning) on in-vivo multispectral fluorescence lifetime imaging microscopy data for oral carcinoma detection (Photochem. Photobiol. 2016).
- ## TEACHING & MENTORING
- **Texas A&M University, Department of Electrical Engineering** College Station, TX, USA
Grader (Signals and Systems) Jan 2014 – Apr 2014
 - Designed grading rubrics and graded coding, written assignments, and exams.
 - **GEM Center, The University of Texas at Dallas** Richardson, TX, USA
Math Tutor Jan 2011 – May 2011
 - Tutored undergraduate-level mathematics and physics.

Selected Publications

- **J. Lee**, C. Kim, H. Kim, K. Lee, J. Lee. “Soft Equivariance Regularization for Invariant Self-Supervised Learning.” *ICLR 2026 (accepted)*. [pdf]
- **J. Lee**, H. Nam, K. Lee, S. Hahn. “Compact and De-Biased Negative Instance Embedding for Multi-Instance Learning on Whole-Slide Image Classification.” *ICASSP 2024*. [pdf]
- K. Lee, S. Lee, H. Hyun, S. Hahn, E. Choi, **J. Lee**. “Learning Missing Modal Electronic Health Records with Unified Multi-modal Data Embedding and Modality-Aware Attention.” *MLHC 2023*. **Spotlight**. (Co-first; corresponding author.) [pdf]
- **J. Lee**, J. E. Oh, I. Shin, Y. S. Kim, D. K. Sohn, T. S. Kim, I. S. Kweon. “Moving from 2D to 3D: volumetric medical image classification for rectal cancer staging.” *MICCAI 2022*. [pdf] [video]

Workshop

- K. Lee, J. Won, H. Hyun, S. Hahn, E. Choi, **J. Lee**. “Self-Supervised Predictive Coding with Multimodal Fusion for Patient Deterioration Prediction in Fine-grained Time Resolution.” *ICLR 2023 Trustworthy ML for Healthcare Workshop*. Oral Presentation. **Best Paper Honorable Mention**. (Corresponding author.) [pdf]

Journal

- **J. Lee**, J. E. Oh, M. J. Kim, B. Y. Hur, D. K. Sohn. “Reducing the Model Variance of Rectal Cancer Segmentation Network.” *IEEE Access*, 2019. [pdf]
- H. Yoon, **J. Lee**, J. E. Oh, H. R. Kim, S. Lee, H. J. Chang, D. K. Sohn. “Tumor Identification in Colorectal Histology Images Using a Convolutional Neural Network.” *Journal of Digital Imaging*, 2018. [pdf]
- B. H. Malik, **J. Lee**, S. Cheng, R. Cuenca, J. M. Jabbour, Y.-S. L. Cheng, J. M. Wright, B. Ahmed, K. C. Maitland, J. A. Jo. “Objective Detection of Oral Carcinoma with Multispectral Fluorescence Lifetime Imaging *In Vivo*.” *Photochemistry and Photobiology*, 2016. (Co-first author.) [pdf]

In Preparation

- **J. Lee**, C. Kim, K. Lee. “Decoupling Equivariance from Invariant Self-Supervised Learning via Null-Space Projection.” In preparation.
- C. Kim, **J. Lee**, K. Lee, D. Yoon, E. Yang. “SPAM: Sampling Pattern Meta-Learning for Domain Generalization on Irregular Time Series.” In preparation.

HONORS & RECOGNITION

- **MLHC 2023**: Spotlight paper (Learning Missing Modal EHR with modality-aware attention).
- **ICLR 2023 TML4H Workshop**: Best Paper Honorable Mention; Oral Presentation.

REFERENCES

- **Prof. Eunho Yang** (Research supervisor)
Associate Professor, Kim Jaechul Graduate School of AI, KAIST; Email: eunhoy@kaist.ac.kr
- **Prof. Juho Lee** (Research collaborator)
Associate Professor, Kim Jaechul Graduate School of AI, KAIST; Email: juholee@kaist.ac.kr
- **Prof. Dae Kyung Sohn** (Research Supervisor)
Professor, Dept of Public Health & Artificial Intelligence, National Cancer Center; Email: gsgsbal@ncc.re.kr